

```
In [1]: import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn import metrics

import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: data = pd.read_csv("logistic-regression-dataset-2.csv")

print(data.shape)
data.head()
```

(36, 4)

```
Out[2]:
```

	Disease	Age	Gender	Smoker_status
0	diseased	43	Male	Smoker
1	not diseased	18	Male	Smoker
2	diseased	22	Female	Non-smoker
3	diseased	25	Male	Non-smoker
4	not diseased	45	Female	Smoker

```
In [3]: data.isnull().sum()
```

```
Out[3]: Disease      0
Age      0
Gender    0
Smoker_status  0
dtype: int64
```

```
In [8]: data['Disease'] = data['Disease'].map({
    'not diseased': 0,
    'diseased': 1
})

data['Gender'] = data['Gender'].map({
    'Female': 0,
    'Male': 1
})

data['Smoker_status'] = data['Smoker_status'].map({
    'Non-smoker': 0,
    'Smoker': 1
})

data.head()
```

Out[8]:

	Disease	Age	Gender	Smoker_status
0	1	43	NaN	1
1	0	18	NaN	1
2	1	22	NaN	0
3	1	25	NaN	0
4	0	45	NaN	1

In [9]: `feature_cols = ['Age', 'Gender', 'Smoker_status']`

```
X = data[feature_cols]
y = data['Disease']
```

In [10]: `X_train, X_test, y_train, y_test = train_test_split(
 X, y,
 test_size=0.2,
 random_state=5
)`

```
display(
    X_train.shape,
    y_train.shape,
    X_test.shape,
    y_test.shape
)
```

(28, 3)

(28,)

(8, 3)

(8,)

In [20]: `model = LogisticRegression(
 solver='lbfgs',
 max_iter=1000
)`

```
model.fit(X_train, y_train)
```

Out[20]: **LogisticRegression** ⓘ ?

► Parameters

► Fitted attributes

In [21]: `y_pred = model.predict(X_test)
print(y_pred)`

[1 0 0 1 1 0 1 1]

In [22]: `conf_mat = metrics.confusion_matrix(y_test, y_pred)`

```
print('Confusion Matrix :\n', conf_mat)
```

```

Accuracy_score = metrics.accuracy_score(y_test, y_pred)

print('Accuracy Score :', Accuracy_score)

print(
    'Accuracy in Percentage :',
    int(Accuracy_score * 100),
    '%'
)

```

Confusion Matrix :

```
[[2 0]
 [1 5]]
```

Accuracy Score : 0.875

Accuracy in Percentage : 87 %

```
In [23]: print('Accuracy in Percentage :', Accuracy_score * 100, '%')
```

Accuracy in Percentage : 87.5 %

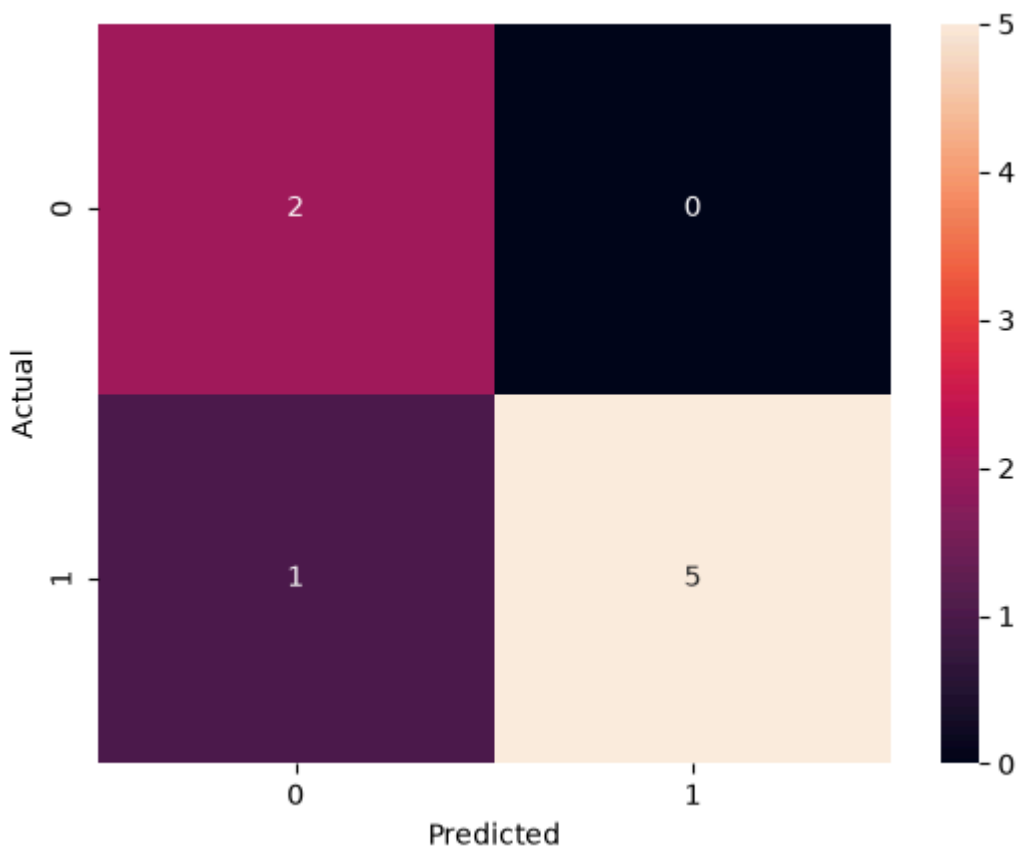
```

In [25]: conf_mat = pd.crosstab(
    y_test,
    y_pred,
    rownames=['Actual'],
    colnames=['Predicted']
)

sns.heatmap(conf_mat, annot=True)

```

Out[25]: <Axes: xlabel='Predicted', ylabel='Actual'>



```

In [26]: print(
    metrics.classification_report(
        y_test,

```

y_pred					
		precision	recall	f1-score	support
0		0.67	1.00	0.80	2
1		1.00	0.83	0.91	6
accuracy				0.88	8
macro avg		0.83	0.92	0.85	8
weighted avg		0.92	0.88	0.88	8